



Boosting Food Production Efficiency

How Mettler Toledo Halogen Moisture Analysis Enables Faster QC, Lean Changeovers, and Quicker Batch Release

Moisture content affects the quality of many food products, which must retain their properties up until the time they are consumed. For example, in snack production, moisture content impacts product texture, taste and shelf life, which means it helps to determine optimum storage conditions. However, moisture content determination has historically been a time-consuming process.

This application note describes how considering the lean method of Single-Minute Exchange of Die (SMED), along with other lean concepts in the moisture analysis process, can contribute to more efficient QC testing and faster batch release to help optimize production.

Keywords or phrases:

Halogen Moisture Analysis (HMA), Single-Minute Exchange of Die (SMED), food quality control, loss-on-drying (LOD), lean manufacturing, batch release optimization, rapid changeover strategies in QC testing, accelerating batch approval with halogen moisture analysis

INTRODUCTION

Generally, as a part of a continual improvement process (CIP or CI), SMED incorporates efforts to reduce the time it takes to complete equipment setup or changeover. The concept is to “externalize” as many changeover steps as possible — that is, make it possible for them to be performed while equipment is running.

This serves to:

- ▶ Reduce downtime
- ▶ Simplify changeover processes
- ▶ Optimize productivity

The original goal of SMED was to reduce changeover time to single digits, or less than 10 minutes. After “lean” principles such as CIP and SMED had been applied for the first time, tool

changeover times went from hours in the 1950s to minutes in the 1990s.

Today, “zero changeover (a kind of super SMED)” is even more dynamic, with changeover times reduced to less than three minutes. In food processing, this kind of rapid changeover has proven key to reducing the time it takes to approve production batches and therefore improve throughput.

For moisture analysis, loss-on-drying (LOD) in a drying oven is the typical reference method noted in food regulations. However, a halogen moisture analyser such as the HX204 from METTLER TOLEDO — which uses halogen moisture analysis (HMA) and is based on the LOD principle — can provide much faster and cost-effective moisture determination. In fact, results are typically delivered in minutes instead of hours.

Using HMA, the thermogravimetric principle is applied: a sample is heated by absorption of infrared radiation from halogen lamp. While the heat source is different, the principle is similar. Hence, the results from the moisture analyser and the drying oven are comparable.

However, HMA, can match the drying-oven reference method in minutes.

Depending on the sample, a moisture analyser requires just 3 to 15 minutes. Moisture determination using a halogen moisture analyser can be even faster with a unique and innovative function known as QuickPredict.

In just two minutes, METTLER TOLEDO QuickPredict predicts the expected measurement result. This allows the laboratory or production manager to decide at a much earlier stage what further steps need to be taken if an OOE (Out-of-Expectation) result is predicted.

When applied to moisture analysis, a faster setup and changeover of samples significantly reduce time to result. Overall, fast moisture analysis speeds up sample processing, which in turn reduces production loss and output variability.

IMPLEMENTING SMED IN FOOD PRODUCTION AND QUALITY CONTROL

Reducing changeover with SMED in any production process requires thorough analysis of the given process. Generally, this will involve the following actions:

- ▶ Observe the current process.
- ▶ Separate the external and internal production steps.
- ▶ Internal activities are those performed when the process is stopped. External activities can be done while production is ongoing.
- ▶ Convert as many internal processes as possible to external processes.
- ▶ Simplify remaining internal activities.
- ▶ Document the new process.
- ▶ Review new method results to see if productivity gains have been made.

SUPPORTING SMED USING HMA

One snack-food company sought faster, higher-quality moisture content results in batch and sample control at the line based on thermogravimetric measurement.

The analyser would be used by shift workers instead of trained lab technicians. This required equipment that was robust and easy-to-use.

Could the HX204 provide the required control in much less time? And could the sampling process and cleanup be reduced to single digits (less than 9 minutes 59 seconds)?

It turns out, the HX204 could reduce the sampling process to meet SMED standards, and even reduced sample changeover to super-SMED status, as described in the following section.

ANALYSING SMED IN-PROCESS

Using the HX204, the production-line sampling process at the company in question involved the following seven steps:

- ▶ The user takes the sample (e.g. potato chips).
- ▶ The sample is prepared (crunched, homogenized).
- ▶ The user logs into the operator's profile. Security levels can also be set, so the operator is only able to change the settings required for his or her profile.



- ▶ The method is started with a pre-defined shortcut. This company uses A and B buttons for method saving. Behind each letter is a pre-defined method e.g. for crisp and cereals.
- ▶ The sample is applied onto the pan and weighed. It is important to always use the same sample weight (e.g. 5 g $\pm 10\%$). HX204 offers a weighing aid feature that, when activated, ensures that the operator cannot start the measurement if the sample weight is out of tolerance. Furthermore, with HX204, this is an ergonomic process because of the moisture analyzer stand. (Operators tend to suffer from problems caused by poorly designed devices. Providing comfort during work is an aspect of CIP.)
- ▶ The measurement is in progress.
- ▶ The result is displayed. Depending on the method and sample characteristics, results will be showed on the terminal in less than 10 minutes. Result documentation can be done via network printer or stored directly as a PDF-A file fulfilling the highest security requirements, such as those of the FDA².



The measured sample and pan are removed, and a new pan is added, making the equipment ready for the next sample almost immediately. In case of spills, they can be quickly cleaned/wiped off the flat stainless-steel surfaces. The unit's construction reduces sample changeover time to less than 30 seconds (well under the required 9:59 for SMED and exceeding zero changeover).

The results of the super-SMED process can be presented in terms of ROI using the following calculation:

Before, SMED (as found) used to take 11 minutes, 3 times per day (3 shifts 6 days per week), 200 working days per year. In total non-added value time: 6,600 minutes per year or 110 hours were lost to the process.

SMED (as left) takes 30 seconds $\times$ 3 shifts, 6 days per week or 200 working days per year. The zero-changeover process required just 300 minutes or 5 hours over the course of the year.

The benefits of the changes over a year are as follows:

- ▶ 105 hours saved
- ▶ Fewer non-added value steps thanks to implementing KPI (Key Performance Indicators) and SMED to reduce sample setup and changeover times for each device.
- ▶ A process (SMED) that can be applied to all the other devices in their manufacturing plants and laboratories to create additional time, effort, and cost savings.

The device can also be tested for accuracy. With the in-built test procedure and SmartCal, a Moisture Analyzer Test Substance, an operator can easily test the device in 10 minutes.

SmartCal adheres to principles of Good Manufacturing Practice (GMP), Good Laboratory Practices (GLP) and ISO as a test substance certified in Bundesanstalt für Materialforschung und -prüfung (BAM), a Germany-based Testing Laboratory.



Moreover, EasyDirect™ Moisture Software enables the simplification and optimization of moisture analyzer data management by collecting data automatically, creating reports, visualizing results and more (www.mt.com/moisture-software).

ADVANTAGES OF SMED IMPLEMENTATION

Applying SMED to processes in your food production facility such as moisture determination, when combined with the latest technology, can bring the following benefits:

- ▶ **Accurate results.** In this instance, advanced halogen technology proved key to fast heating and precise temperature control.
- ▶ **Better use of bench space.** The device is small, barely the size of a piece of paper, hence optimizing lab space (another aspect of lean processing). The stand reduces the footprint further and enables ergonomic operation.
- ▶ **Shorter production time.** The reduction of moisture determination from hours to minutes, as well as a 30-second sample changeover and cleaning procedure,

makes moisture determination an easy, repeatable part of batch release.

- ▶ **Fast initiation.** The graphical One Click™ user interface significantly reduces instrument training time and also allows methods to be started with one motion.
- ▶ **Better process control.** Smart graphics including a real-time drying curve visualize the ongoing measurement on the display. The user can control charts and trend moisture content over time.
- ▶ **Improved safety and quality.** With password protected user profiles, safety is ensured for fully regulated operations. HX204 offers simpler setup, easier method-initiation.
- ▶ **User-friendly operation.** An ergonomic design helps to ensure operator safety and eliminate repetitive stress injuries.

- ▶ **Fast cleaning.** The flat surface of the weighing pan is completely smooth and sealed. Cleaning a moisture analyzer has never been as easy and fast. Spills are contained in the draft shield and can be removed before the next measurement. The position of the display on the stand also helps to reduce the risk of getting the terminal dirty and protects it from damage caused by heavy-duty use.
- ▶ **Regulatory compliance.** Password-protected profiles, secure database management and more help to be compliant with regulations.
- ▶ **Data-collection ease.** Software featuring the EasyDirect functionality allows the connection of up to five devices to collect results in a central database.

It also provides excellent handling for shift workers and experienced lab technicians alike.



Halogen Moisture Analyzer HX204